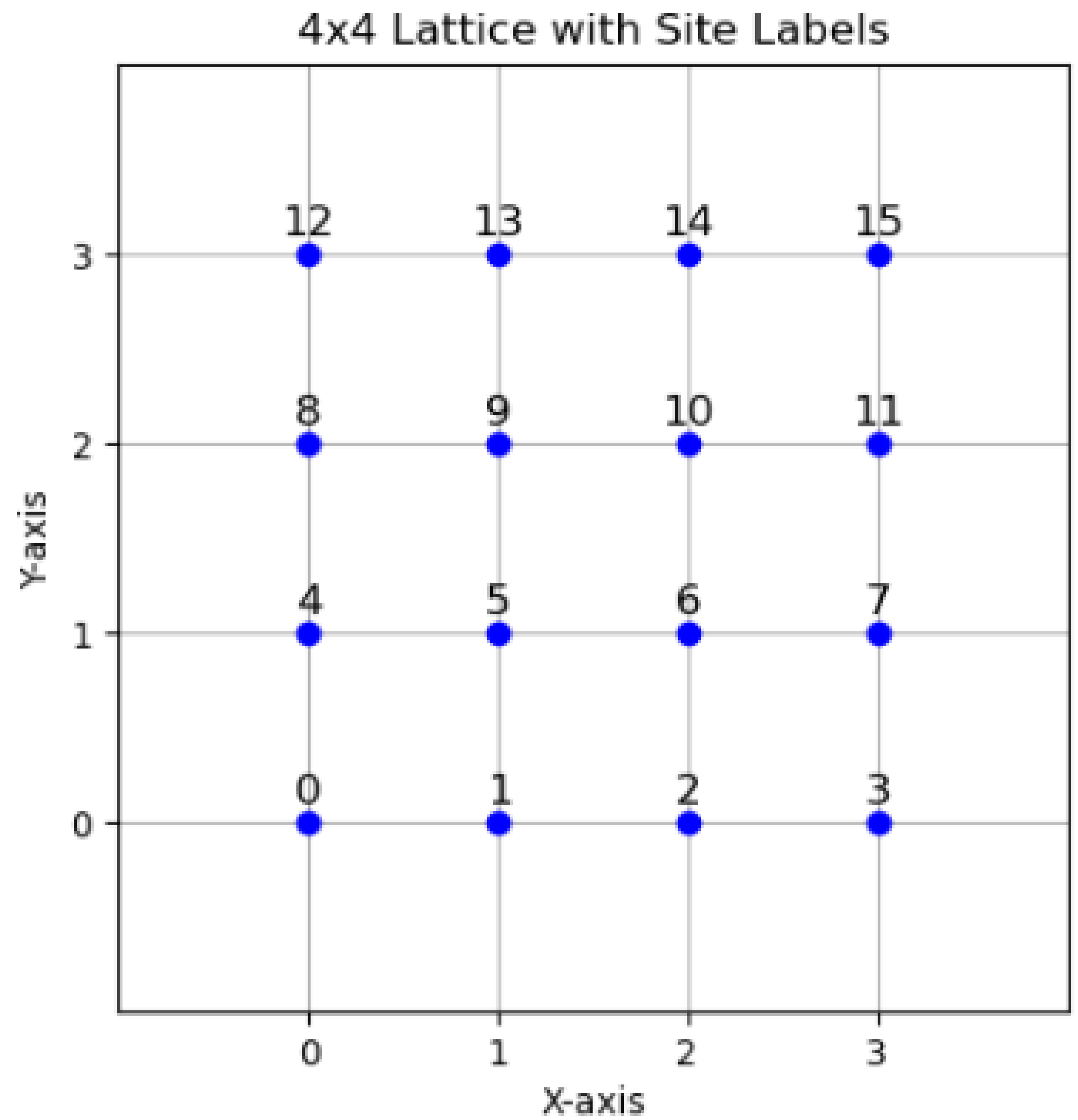

First Principles Solutions of a Geometrically Frustrated Quantum Spin System: The Shastry-Sutherland Model

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10-11 AM, Monday, Nov. 25, 2024, Physics 298, Faculty Lounge

What is Geometrical Frustration?

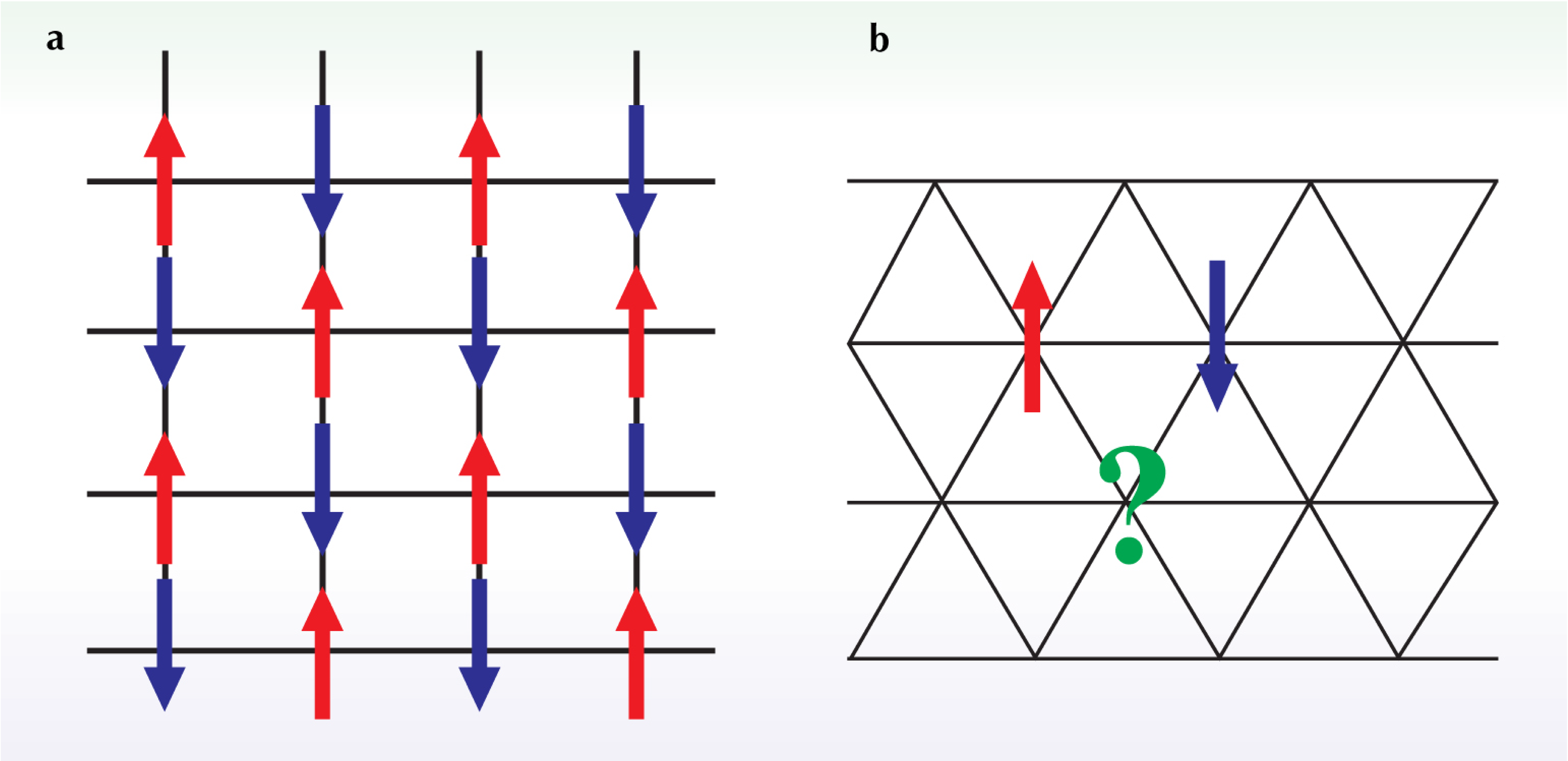

Figure 1. <u>*A geometrically frustrated system*</u> is one in which the geometry of the lattice precludes the simultaneous minimization of all interactions. (**a**) In the un-frustrated antiferromagnet on the square lattice, each spin can be anti-aligned with all its neighbors. (**b**) On a triangular lattice, such a configuration is impossible: Three neighboring spins cannot be pairwise anti-aligned, and the system is frustrated. (*Adapted from Physics Today. 2006;59(2):24-29. doi:10.1063/1.2186278*)

Introduction: The Shastry-Sutherland Model

$$H^{\mathrm{ssm}} = J' \sum_{n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + J \sum_{n.n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

•**Type**: Geometrically frustrated 2D quantum spin model.

•**Lattice**: Square lattice with alternating diagonal dimer bonds.

•**Interactions**:
- •J': Nearest-neighbor (n.n.) antiferromagnetic interaction.
- •J: Stronger next-nearest-neighbor (n.n.n.) diagonal dimer interaction.

•**Frustration**:
- •Competing J' and J interactions prevent simple spin ordering.
- •Results in non-degenerate ground state.

•**Ground State**: Exactly solved for *J'/J <= 0.5* by Shastry and Sutherland (1981)
- •*Dimerized phase*: Singlet pairs form along diagonal bonds with J coupling strength.
- •*Magnetization plateaus*: Observed under external magnetic fields, indicating exotic phases.
- •*Applications*: Explains behavior of materials like $SrCu_2(BO_3)_2$ and many others with dimerized quantum magnetism.

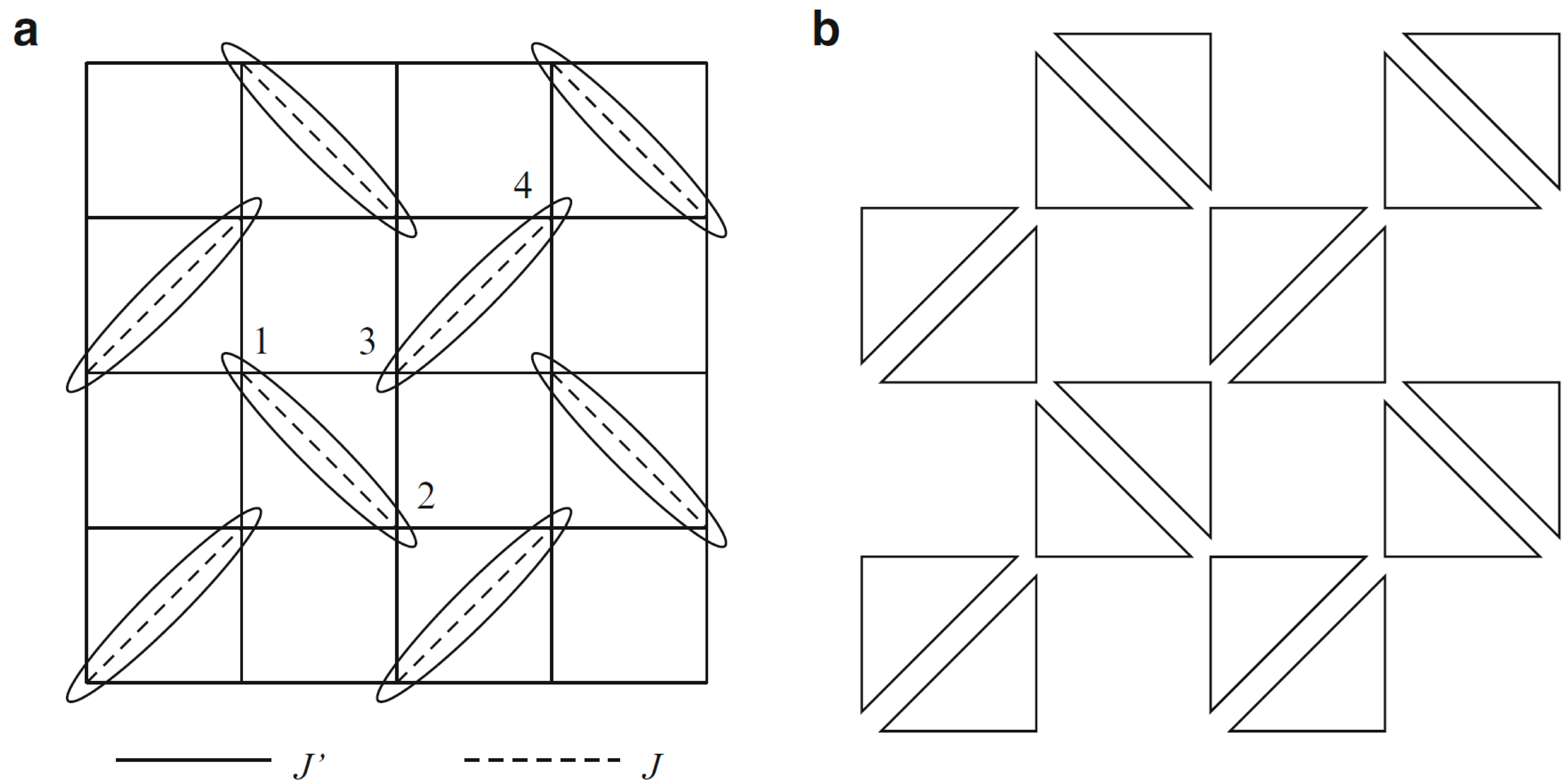

Figure 2. (**a**) The two-dimensional spin-1/2 Shastry–Sutherland Heisenberg model, where an exact dimer-product ground state is realized. Ellipses represent the dimer singlet bonds. (**b**) For *J'/J <= 0.5*, the Hamiltonian is described as a sum of triangular clusters. *(Adapted from Introduction to Frustrated Magnetism: Materials, Experiments, Theory. (2011). Germany: Springer Berlin Heidelberg.)*

Research Motivation and Objectives

$$H^{\text{ssm}} = J' \sum_{n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + J \sum_{n.n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

Motivation

• Ground state analytically solved for $J'/J \leq 0.5$; unknown for larger J'/J.

• Experimental evidence shows a phase transition:
 •**Dimerized state → Néel ordered state**.

• Controversy over intermediate phase near $J'/J \sim 0.7$ (Miyahara and Ueda, 2003) :

1. Helical ordered state (Albrecht and Mila, 1996; Chung et al., 2001).
2. Plaquette Resonating Valence Bond (RVB) state (Koga and Kawakami, 2000; Carpentier and Balents, 2002; Lauchli et al., 2002).

Objectives

• Develop an **Exact Diagonalization (ED)** algorithm for a 4x4 (16-site) Shastry-Sutherland lattice.

• Explore how the ground state changes as J'/J varies.

• Try to understand if there are hints of the proposed intermediate phases (phase transitions) even in our small 4x4 lattice.

Exact Diagonalization: Main Idea

$$H^{\mathrm{ssm}} = J' \sum_{n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + J \sum_{n.n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

Solve an eigenvalue problem for a finite quantum many body system, by numerical diagonalization of H with no approximations.

$$H|\psi\rangle = E|\psi\rangle$$

Exact Diagonalization: Ingredients

Solve an eigenvalue problem for a finite quantum many body system, by numerical diagonalization of H with no approximations.

$$H|\psi\rangle = E|\psi\rangle$$

Ingredients

1. A numerical representation of the basis states in the **Hilbert Space** H.
2. A numerical representation of the **Hamiltonian matrix** H^{ssm}.
3. A **diagonalization method** to calculate the desired eigenvalues and eigenvectors.
4. Understand the **consequences of the spin-symmetry** on the eigenvalue spectrum.
5. A set of **observables** whose expectation values are calculated.

Step 1: Basis Construction

- Basis states of the **Hilbert Space** *H* need to be represented in a computer.

- We need a simple representation that makes it easy to act with the Hamiltonian or other operators.

- In the Shastry-Sutherland model, the lattice sites are occupied by spin-1/2 particles. Hence, we can use the binary representation: **1** for spin-up and **0** for spin-down.

- For a 16-site problem like ours, we have a 16-index ket representation:

  $$|\, s_1 s_2 s_3 \ldots s_{16}\rangle \text{ for each } |\, s_i\rangle = \{0,1\}.$$

- Since each site has two possibilities, there are 2^{16} states! Note that the Hilbert space grows *exponentially* as the no. of sites increases. This will factor into computational time complexity.

Step 1: Basis Construction - $S_z = 0$

- Since the Hilbert space is huge, we are motivated to look for symmetries that allow us to obtain the ground state after some dimensionality reduction.

- Shastry and Sutherland in their foundational paper (1981) show that the exactly known ground state is the "product of dimer singlets." This gives us a nice strategy for reducing the Hilbert space dimension.

- We need to care, for the time being, about the singlets only! Hence, we resort to $S_z = 0$ regime as singlets are, by definition, states with $|\ s = 0, m = 0\rangle$.

- This reduces our Hilbert space quite significantly; it brings it down from 2^{16} to:

$$^{16}C_8 = \frac{16!}{8!\,8!} = 12870.$$

- With the restriction of $S_z = 0$, we generate the 12870 "filtered" basis states by choosing all permutations of eight 1s and eight 0s in our 16-index ket.

Digression: Theory of Addition of Spins

- Using a recursive code and theory of addition of spins, we compute all possible spins and their occurrences for our 16-spin ½ lattice problem. This is the first thing we did to explore what we could do for dimensionality reduction. Luckily, this served as a check later on…

```
Possible Spins and Occurrences:
Spin: 0.0, Occurrence: 1430
Spin: 1.0, Occurrence: 3432
Spin: 2.0, Occurrence: 3640
Spin: 3.0, Occurrence: 2548
Spin: 4.0, Occurrence: 1260
Spin: 5.0, Occurrence: 440
Spin: 6.0, Occurrence: 104
Spin: 7.0, Occurrence: 15
Spin: 8.0, Occurrence: 1

Final Spins and Degeneracies:
Spin      Degeneracy  Degeneracy per occurrence
0.0       1430.0      1.0
1.0       10296.0     3.0
2.0       18200.0     5.0
3.0       17836.0     7.0
4.0       11340.0     9.0
5.0       4840.0      11.0
6.0       1352.0      13.0
7.0       225.0       15.0
8.0       17.0        17.0

Dimension of Hilbert space: 65536
Sum of degeneracies: 65536.0
Degeneracies sum up correctly.
Number of singlets (J=0): 1430
```

Step 2: Hamiltonian Construction

$$H^{\text{ssm}} = J' \sum_{n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + J \sum_{n.n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

- There are two kinds of interactions in our Hamiltonian, and we construct them separately, and add them, i.e., $H^{ssm} = H_{nn} + H_{nnn}$.

- We go back to the fundamentals and bring back results from two spin interactions by $H_{ij} = S_i \cdot S_j$.

Step 2: Hamiltonian Construction

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Basis States:

1. $|\uparrow\uparrow\rangle = |+\frac{1}{2}, +\frac{1}{2}\rangle$
2. $|\uparrow\downarrow\rangle = |+\frac{1}{2}, -\frac{1}{2}\rangle$
3. $|\downarrow\uparrow\rangle = |-\frac{1}{2}, +\frac{1}{2}\rangle$
4. $|\downarrow\downarrow\rangle = |-\frac{1}{2}, -\frac{1}{2}\rangle$

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \tfrac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)$$

$$\mathbf{S}_i \cdot \mathbf{S}_j |\uparrow\uparrow\rangle = \frac{1}{4} |\uparrow\uparrow\rangle$$

$$\mathbf{S}_i \cdot \mathbf{S}_j |\downarrow\downarrow\rangle = \frac{1}{4} |\downarrow\downarrow\rangle$$

$$\mathbf{S}_i \cdot \mathbf{S}_j |\uparrow\downarrow\rangle = -\frac{1}{4} |\uparrow\downarrow\rangle + \frac{1}{2} |\downarrow\uparrow\rangle$$

$$\mathbf{S}_i \cdot \mathbf{S}_j |\downarrow\uparrow\rangle = -\frac{1}{4} |\downarrow\uparrow\rangle + \frac{1}{2} |\uparrow\downarrow\rangle$$

Step 2: Hamiltonian Construction

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)$$

$$\mathbf{S}_i \cdot \mathbf{S}_j | \uparrow\uparrow\rangle = \frac{1}{4} | \uparrow\uparrow\rangle$$

$$\mathbf{S}_i \cdot \mathbf{S}_j | \downarrow\downarrow\rangle = \frac{1}{4} | \downarrow\downarrow\rangle$$

$$\mathbf{S}_i \cdot \mathbf{S}_j | \uparrow\downarrow\rangle = -\frac{1}{4} | \uparrow\downarrow\rangle + \frac{1}{2} | \downarrow\uparrow\rangle$$

$$\mathbf{S}_i \cdot \mathbf{S}_j | \downarrow\uparrow\rangle = -\frac{1}{4} | \downarrow\uparrow\rangle + \frac{1}{2} | \uparrow\downarrow\rangle$$

- This gives us complete action of spin-spin interaction term in our Hamiltonian. We just need to choose the index $\{i,j\}$ wisely and assign coupling strengths *J and J'* to get the full Hamiltonian.

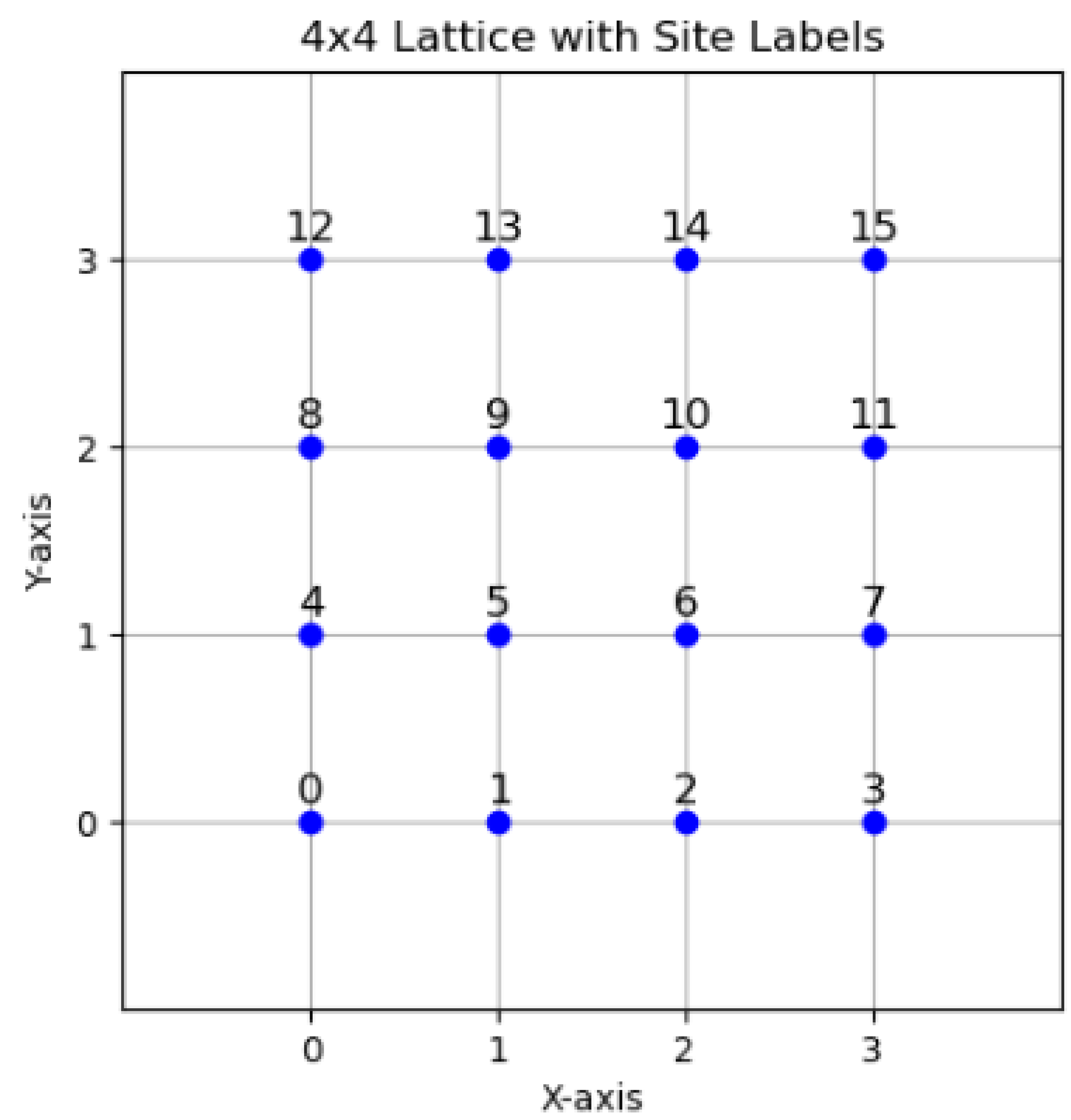
- Here comes another assumption: **periodic boundary conditions**. This has been used in literature in many theoretical studies before and have been seen to reasonably agree with experimental realizations of the model. It also allows us to well analyze the limited 16-site problem we are trying to solve.

Step 2: Hamiltonian Construction - H_{nn}

- We now introduce a labeling convention for all our sites for appropriate indexing as shown in the plot on the right.

- For the nearest neighbour interaction, H_{nn}, we choose a site i and then, the pick $(i+1)$ on the right and $(i+p)$ up.

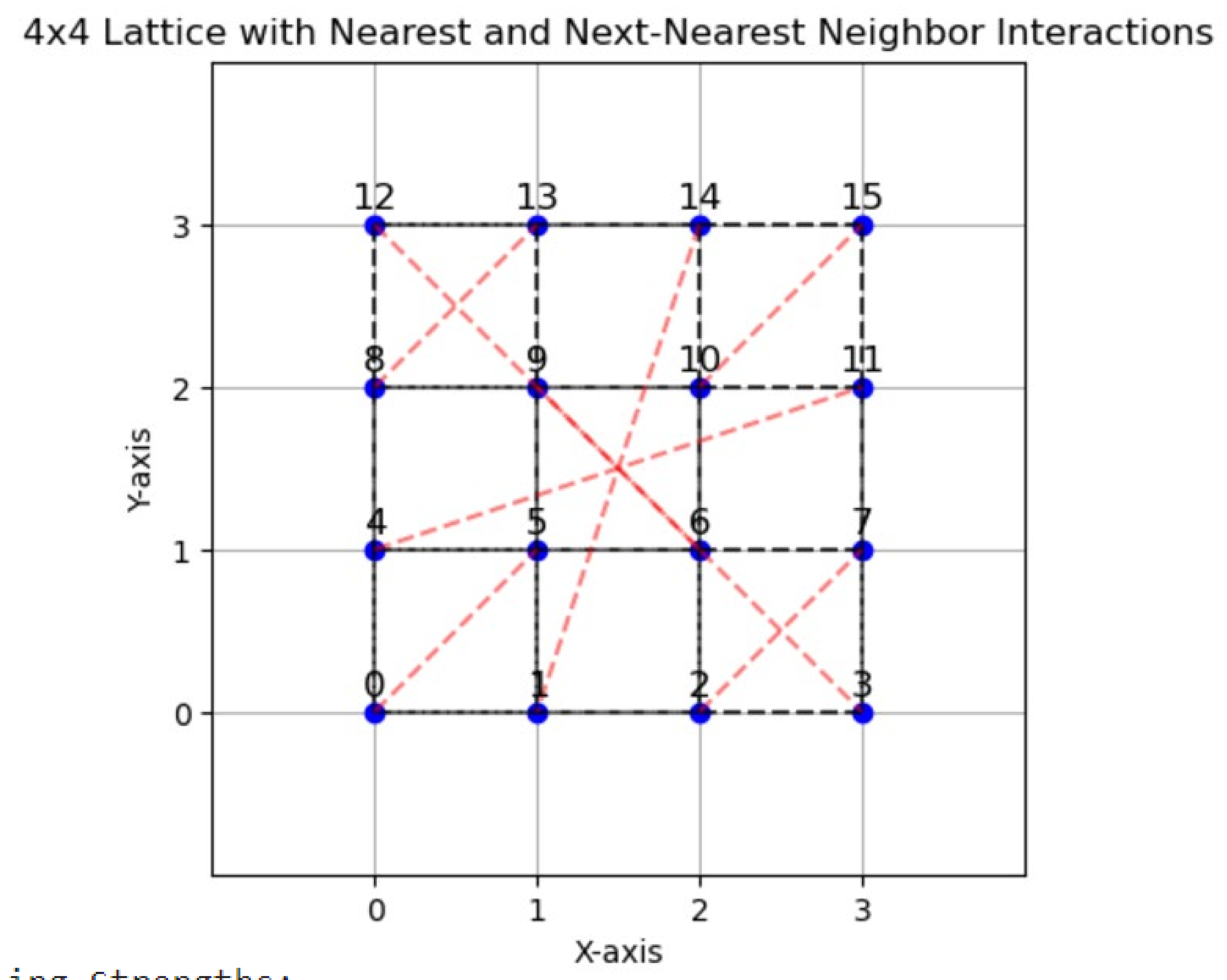
```
p = 4  # Periodicity in x-direction

# List to store nearest-neighbor pairs
nearest_neighbors = []
for i in range(n_spins):
    # Couple with the spin to the right
    right = (i + 1) if (i + 1) % p != 0 else i + 1 - p
    nearest_neighbors.append((i, right))
    # Couple with the spin above
    up = (i + p) % n_spins
    nearest_neighbors.append((i, up))
```



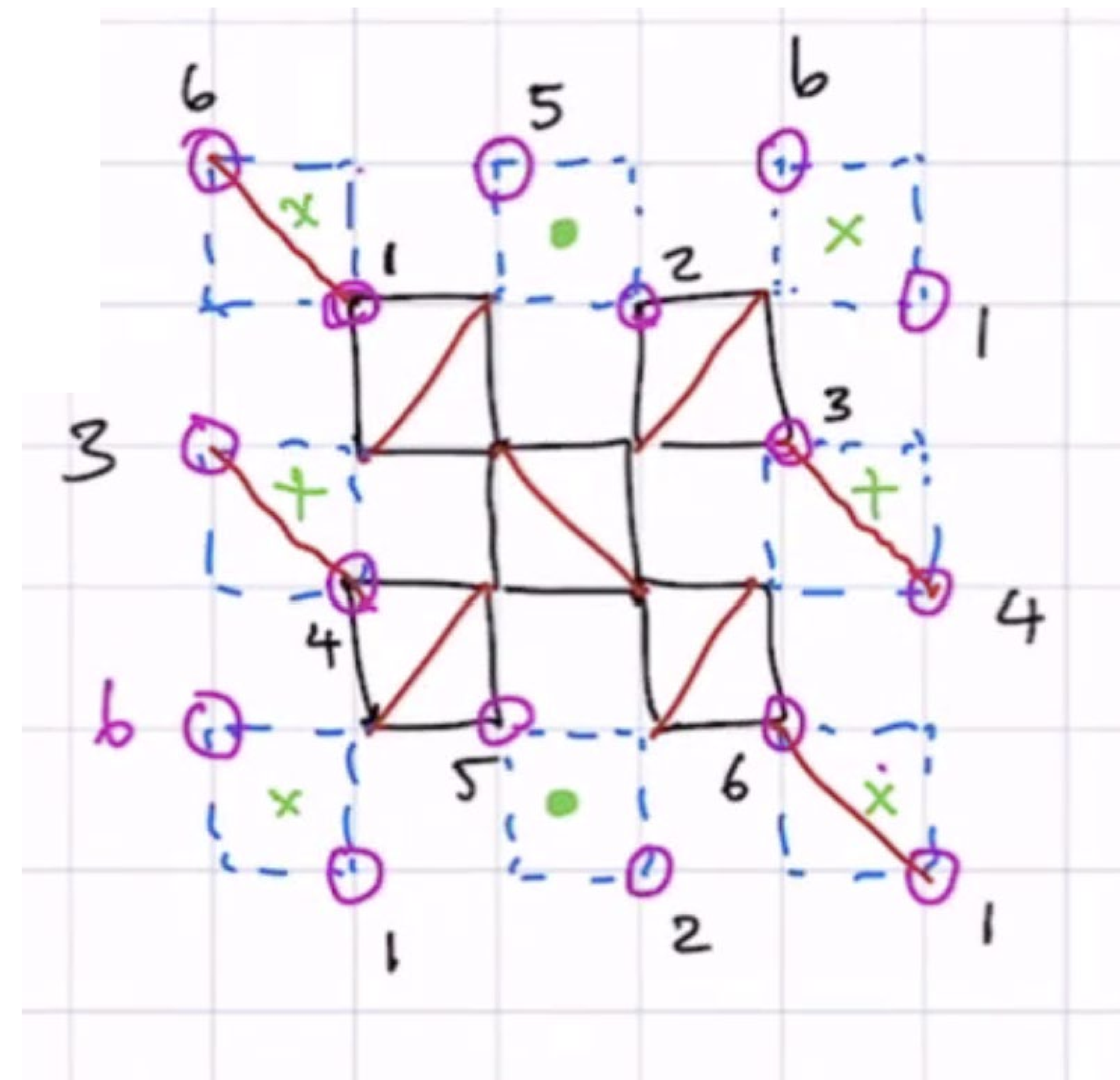
Step 2: Hamiltonian Construction - H_{nnn}

- For the next-nearest neighbour interaction, H_{nnn}, we need to choose the diagonal bonds more appropriately.

```python
# Define next nearest neighbor interactions
J = 1
p = 4
next_nearest_neighbors = []

for y in range(4):
    for x in range(4):
        k = y * 4 + x
        if x % 2 == 1:
            continue  # Skip odd x to avoid double counting
        if y % 2 == 0:
            # Couple (x, y) to (x+1 mod p, y+1 mod p)
            neighbor_x = (x + 1) % p
            neighbor_y = (y + 1) % p
            j = neighbor_y * 4 + neighbor_x
            next_nearest_neighbors.append((k, j))
        else:
            # Couple (x, y) to (x-1+p mod p, y+1 mod p)
            neighbor_x = (x - 1 + p) % p
            neighbor_y = (y + 1) % p
            j = neighbor_y * 4 + neighbor_x
            next_nearest_neighbors.append((k, j))

# Assign strength J (for now set J = 2)
next_nearest_neighbor_strengths = {pair: J for pair in next_nearest_neighbors}

# Print next nearest neighbors and their coupling strengths
print("Next Nearest Neighbor Pairs and Coupling Strengths:")
for pair, strength in next_nearest_neighbor_strengths.items():
    print(f"Pair {pair}: Strength {strength}")
```




```
Next Nearest Neighbor Pairs and Coupling Strengths:
Pair (0, 5): Strength 1
Pair (2, 7): Strength 1
Pair (4, 11): Strength 1
Pair (6, 9): Strength 1
Pair (8, 13): Strength 1
Pair (10, 15): Strength 1
Pair (12, 3): Strength 1
Pair (14, 1): Strength 1
```



Step 2: Hamiltonian Construction - H^{ssm}

- We know how to act on any two spins with a chosen $\{i,j\}$ as shown previously. In our 16-index ket, this will manifest in the following manner:

Example 1:

```
Initial state:(0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1)
Resulting state: (0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1), Coefficient: 0.25, Matrix Index: H[0, 0]
Non-zero element in H matrix: H[0, 0] = 0.25
```

Example 2:

```
Initial state:(0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1)
Resulting state: (0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0), Coefficient: 0.5, Matrix Index: H[0, 8]
Resulting state: (0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1), Coefficient: -0.25, Matrix Index: H[0, 0]
Non-zero element in H matrix: H[0, 8] = 0.5
Non-zero element in H matrix: H[0, 0] = -0.25
```

- Hence, we do this for all possible $\{i,j\}$ in each interaction case, sum them to get H_{nn} and $\mathrm{H_{nnn}}$, which then gives us the full Hamiltonian:

$$\boxed{H^{ssm} = H_{nn} + H_{nnn}}$$

Step 3: Diagonalize the Hamiltonian - H^{SSM}

- This is by-far the easiest step for our dimension reduced case of 12870 basis states in the $S_z = 0$ regime. We can directly use a nice Python library for linear algebra called Numpy.

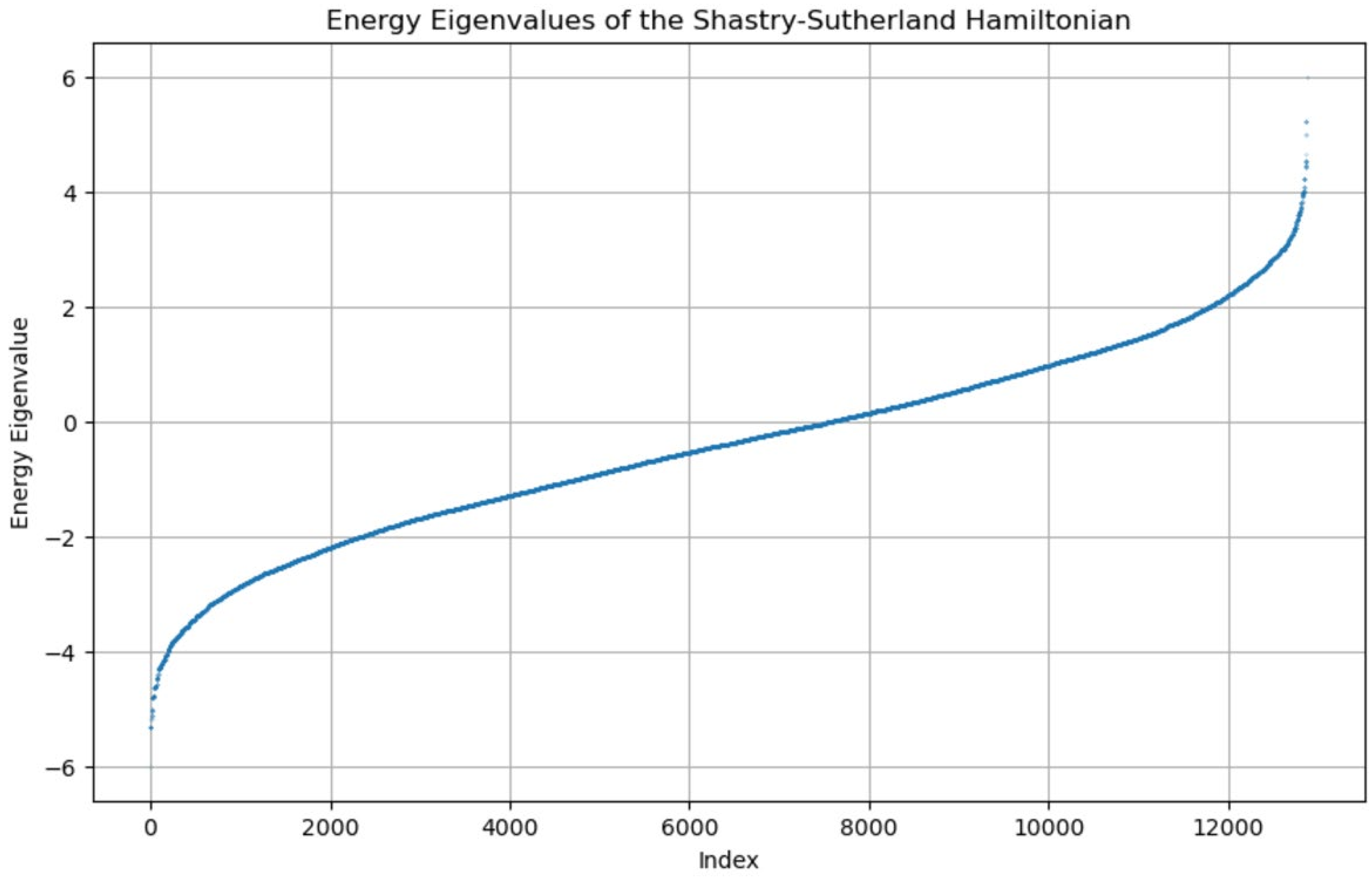
```
import numpy as np

# Calculate eigenvalues and eigenvectors
eigenvalues, eigenvectors = np.linalg.eigh(H_SSM)

#Print the eigenvalues
print("Eigenvalues of the Shastry-Sutherland Hamiltonian matrix:")
print(eigenvalues)
```

- Now, we have all the eigenvalues and eigenvectors of our 12870-dimensional Shastry-Sutherland Hamiltonian. We can check the minimum value to get the ground state energy. In the case of $J'/J \leq 0.5$, Shastry and Sutherland (1981) have shown that the ground state is $-\frac{3J}{8}N_s$; hence, **it depends only the diagonal bond strength *J*** for this parameter range.

- Our numerical calculation matches this analytical result precisely!

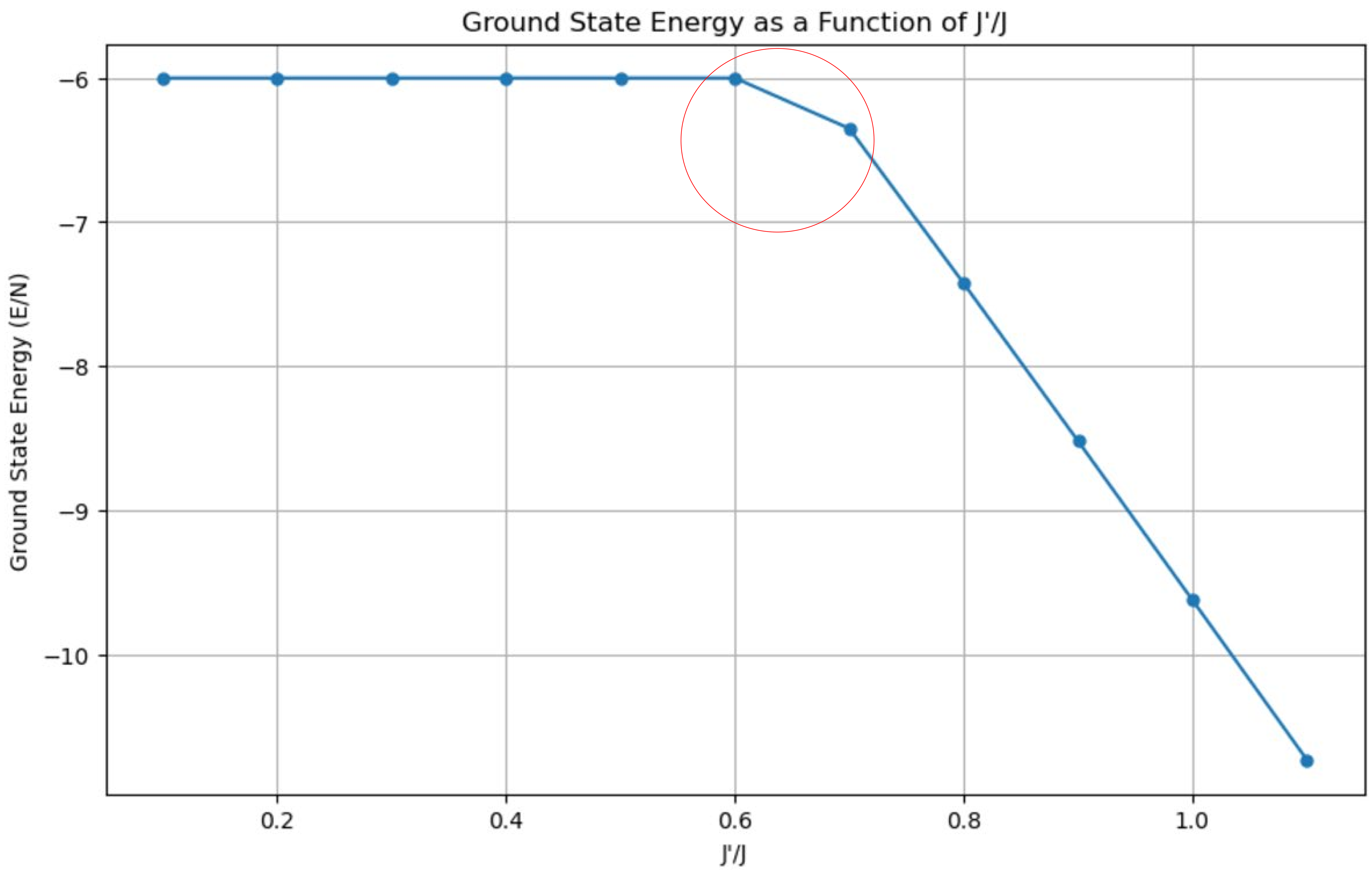
The Energy Spectrum in $S_z = 0$ regime

- We now have the entire energy spectrum of the Hamiltonian for our 16-site problem in the $S_z = 0$ regime.



Digression: What about $J'/J \geq 0.5$?

- Now, since we have a numerical algorithm, we have tools to poke the non-analytic regime for our interaction Hamiltonian. We vary the *J'/J* ratio in our code to see how the ground state energy changes. We do this by setting *J=1* and changing *J'*. Note that the theory result then predicts the ground state energy = $\frac{-3J}{8} N_s = \frac{-3J}{8} 16 = -6J = -6$, matching our numerical one.



Interesting Theory: Need for Excited States!

- It is clear that around $J'/J = 0.7$ the ground state starts deviating distinctly from the dimerized state. This may be related to the quantum phase transition suggested in the literature around ~0.68 for large lattices.

- We now show that the product of dimer singlet ***continues*** to be the eigenstate for all J'/J ratios.

- To do this, we start by taking the state with product of dimer singlets and call it $|\psi_o\rangle$. First, we take the H_{nnn} part and act on $|\psi_o\rangle$. By construction,

$$H_{nnn} \mid \psi_0\rangle = E_0 \mid \psi_0\rangle.$$

- We look at two spins at a time to clarify this:

Interesting Theory: Need for Excited States!

The singlet state is defined as

$$|\mathrm{Singlet}\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right).$$

The operator $\mathbf{S}_i \cdot \mathbf{S}_j$ is expressed as

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+).$$

When $\mathbf{S}_i \cdot \mathbf{S}_j$ acts on the singlet state, the term $S_i^z S_j^z$ gives a contribution of $-\frac{1}{4}$, since S_i^z and S_j^z contribute $\pm\frac{1}{2}$ antisymmetrically. The terms $S_i^+ S_j^-$ and $S_i^- S_j^+$ annihilate the singlet due to its antisymmetric nature. Combining these contributions, the total eigenvalue is $-\frac{3}{4}$. Therefore, the singlet state satisfies the eigenvalue equation:

$$\mathbf{S}_i \cdot \mathbf{S}_j |\mathrm{Singlet}\rangle = -\frac{3}{4} |\mathrm{Singlet}\rangle.$$

Interesting Theory: Need for Excited States!

- Thus, for our 4x4 lattice, we get that:

$$\boxed{E_0 = -\frac{3}{4}\frac{N_s}{2}}$$

as the theory predicates for *J'/J <= 0.5*.

- But we have yet ignored the H_{nn} part. Since we already get the theoretically solved energy, there should not be any extra energy contribution. Then, it must hold that:

$$H_{nn} \mid \psi_0\rangle = 0.$$

- We reason here by considering a single plaquette first and the argument generalizes to all plaquettes.

- Take a plaquette with spin sites labelled 1,2,3,4 in counterclockwise direction. The diagonal dimer singlet is between the spin at site 1 and site 3.

Interesting Theory: Need for Excited States!

The nearest-neighbor Hamiltonian can be rewritten as

$$H_{\mathrm{nn}} = \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3 + \mathbf{S}_3 \cdot \mathbf{S}_4 + \mathbf{S}_4 \cdot \mathbf{S}_1 = (\mathbf{S}_1 + \mathbf{S}_3) \cdot (\mathbf{S}_2 + \mathbf{S}_4).$$

To show $H_{\mathrm{nn}}|\psi_{\mathrm{singlet}_{13}}\rangle = 0$, it suffices to prove $(\mathbf{S}_1 + \mathbf{S}_3)|\psi_{\mathrm{singlet}_{13}}\rangle = 0$. The singlet state $|\psi_{\mathrm{singlet}_{13}}\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow_1\downarrow_3\rangle - |\downarrow_1\uparrow_3\rangle\right)$ has total spin $S = 0$, ensuring that the total spin operator annihilates it. Explicitly,

$$(S_1^z + S_3^z)|\psi_{\mathrm{singlet}_{13}}\rangle = 0, \quad (S_1^\pm + S_3^\pm)|\psi_{\mathrm{singlet}_{13}}\rangle = 0.$$

Thus,

$$(\mathbf{S}_1 + \mathbf{S}_3)|\psi_{\mathrm{singlet}_{13}}\rangle = 0.$$

Substituting this into H_{nn},

$$H_{\mathrm{nn}}|\psi_{\mathrm{singlet}_{13}}\rangle = (\mathbf{S}_1 + \mathbf{S}_3) \cdot (\mathbf{S}_2 + \mathbf{S}_4)|\psi_{\mathrm{singlet}_{13}}\rangle = 0.$$

Therefore, the nearest-neighbor Hamiltonian vanishes when acting on the singlet $|\psi_{\mathrm{singlet}_{13}}\rangle$.

Interesting Theory: Need for Excited States!

- This generalizes for each plaquette and we get the desired result (for *J=1*):

$$H^{ssm} \mid \psi_0\rangle = Hnn \mid \psi_0\rangle + Hnnn \mid \psi_0\rangle = \boxed{-6 \mid \psi_0\rangle}.$$

- Hence, in the energy diagram, the energy of the product of dimer singlet state $\mid \psi_0\rangle$ will appear a horizontal line at *-6*.

- This tells us that some other state is coming down around *J'/J = 0.7* and replacing $\mid \psi_0\rangle$ as the ground state.

- We then ask: what is this state that comes down? Is it a singlet as well of some other kind? Is it a multiplet? We need to understand the ***initially excited states***!

- This motivates us to look for **the total spin *s*** instead of just *m* values that we have had yet in the $S_z = 0;\ m = 0$ regime. This will also help us recover the whole 65,536-dimensional space with (2s+1) degeneracy per spin *s.* We explore this further…

More Nice Theory: Obtaining the Excited States

- To obtain the excited state energies, we will need to go beyond the restricted regime of $S_z = 0$.

- A natural way would be to obtain the total spin *s* for each of the energy eigenstates. Once we get the total spin *s* for each state, we can use the theory of angular momentum and ladder operators to get all associated states.

- For each *s,* the quantum number *m* has values from *–s to +s*, and *m=0* state only occurs once. Hence, we can, in principle, recover the entire 2^{16}-dimensional Hilbert space.

- The first step towards this calculation is to show that $[H^{ssm}, S_{ssm}^2] = 0$, where, for a single spin, S_i^2 is the total spin operator squared, i.e., $S_i^2 = S_x^2 + S_y^2 + S_z^2$. Then,

$$\mathbf{S}_{\mathrm{ssm}} = \sum_{i=1}^{16} \mathbf{S}_i .$$

More Nice Theory: Obtaining the Excited States

- For a 16-spin ½ system, the total spin operator is:

$$\mathbf{S}_{\text{ssm}} = \sum_{i=1}^{16} \mathbf{S}_i.$$

- Expanding the square of above, we get:

$$\mathbf{S}_{\text{ssm}}^2 = \left(\sum_{i=1}^{16} \mathbf{S}_i\right)^2 = \sum_{i=1}^{16} \mathbf{S}_i^2 + 2\sum_{i<j} \mathbf{S}_i \cdot \mathbf{S}_j.$$

- Then, the spin-1/2 particles give $S_i^2 = \frac{1}{2}\left(\frac{1}{2}+1\right) = \frac{3}{4}$. So, in total, summing over 16 spins, we get:

$$\sum_{i=1}^{16} \mathbf{S}_i^2 = 16 \cdot \frac{3}{4} = 12.$$

- Thus, finally, we have:

$$\mathbf{S}_{\text{ssm}}^2 = 12 + 2\sum_{i<j} \mathbf{S}_i \cdot \mathbf{S}_j.$$

More Nice Theory: Obtaining the Excited States

$$\mathbf{S}^2_{\mathrm{ssm}} = 12 + 2\sum_{i<j} \mathbf{S}_i \cdot \mathbf{S}_j.$$

- Since, $\left[S_i \cdot S_j, S_i \cdot S_j\right] = 0$ holds trivially, we get our desired result:

$$\boxed{[H^{ssm}, S^2_{ssm}] = 0.}$$

- Now, we know that both operators can be simultaneously diagonalized. But, we have already diagonalized H^{ssm}! Hence, we can act with S^2_{ssm} on the energy eigenstates to get the matrix elements in the energy eigenstate basis for each degeneracy class as

$$\langle \psi_i \mid S^2_{ssm} \mid \psi_j \rangle.$$

- Once we get all the S^2_{ssm} for each degeneracy class, we can diagonalize them. Then, we can compare the S^2_{ssm}-eigenvalues to obtain the total spin s, by noting the action on an associated eigenvector:

$$S^2_{ssm} \mid \phi \rangle = s(s+1) \mid \phi \rangle.$$

More Nice Theory: Obtaining the Excited States

- First, we look through the energy eigenvalues and eigenstates previously obtained and group them into degeneracy classes in our code. For $J'/J = 0.5$, we get:

```
Ground state eigenvalue: -6.0
Ground state degeneracy: 1
Total number of degeneracy groups: 4454
Total number of states in S_z=0 regime: 12870
Expected number of states in S_z=0 regime: 12870
The total number of states matches the expected value.
Number of non-degenerate (g=1) elements: 857
```

- Then, we get the matrix elements for each block diagonal irreps of S^2_{ssm} and find the total spins s.

- We confirm that we have recovered the entire Hilbert space by checking:

$$\sum_i^{12870}(2s_i + 1) = 2^{16}$$

- To further explore the group-theoretic implications, we count all the irreps:

More Nice Theory: Obtaining the Excited States

- To further explore the group-theoretic implications of the degenerate subspaces of the energy eigenstates, we need to understand the degeneracies shown below:

```
Total number of matrices: 4454
Matrix orders and their counts:
1 x 1: 857 matrices
4 x 4: 2394 matrices
2 x 2: 1197 matrices
3 x 3: 3 matrices
7 x 7: 2 matrices
20 x 20: 1 matrices
```

Check if $\mathrm{g_i} \times n_{g_i} = 12870$ is true: $857 + 4 \times 2394 + 2 \times 1197 + 3 \times 3 + 7 \times 2 + 20 \times 1 = 12870$, holds!

- From the spin-symmetry alone, we don't expect any degeneracy at all. We suspect that these irrep possibilities arise due to a hidden symmetry related to the geometry of our lattice. We are still looking for a symmetry group of our 4x4 lattice with periodic boundary conditions. Without the BC, the symmetry group is the well-known wallpaper group *p4m*. This is a math problem. Yet to thoroughly explore…

A Check for Total Spin Calculations!

- For further verification, we counted the total no. of spins and their occurrences to see if it matches what we had initially found directly from theory of addition of spins. And it does!

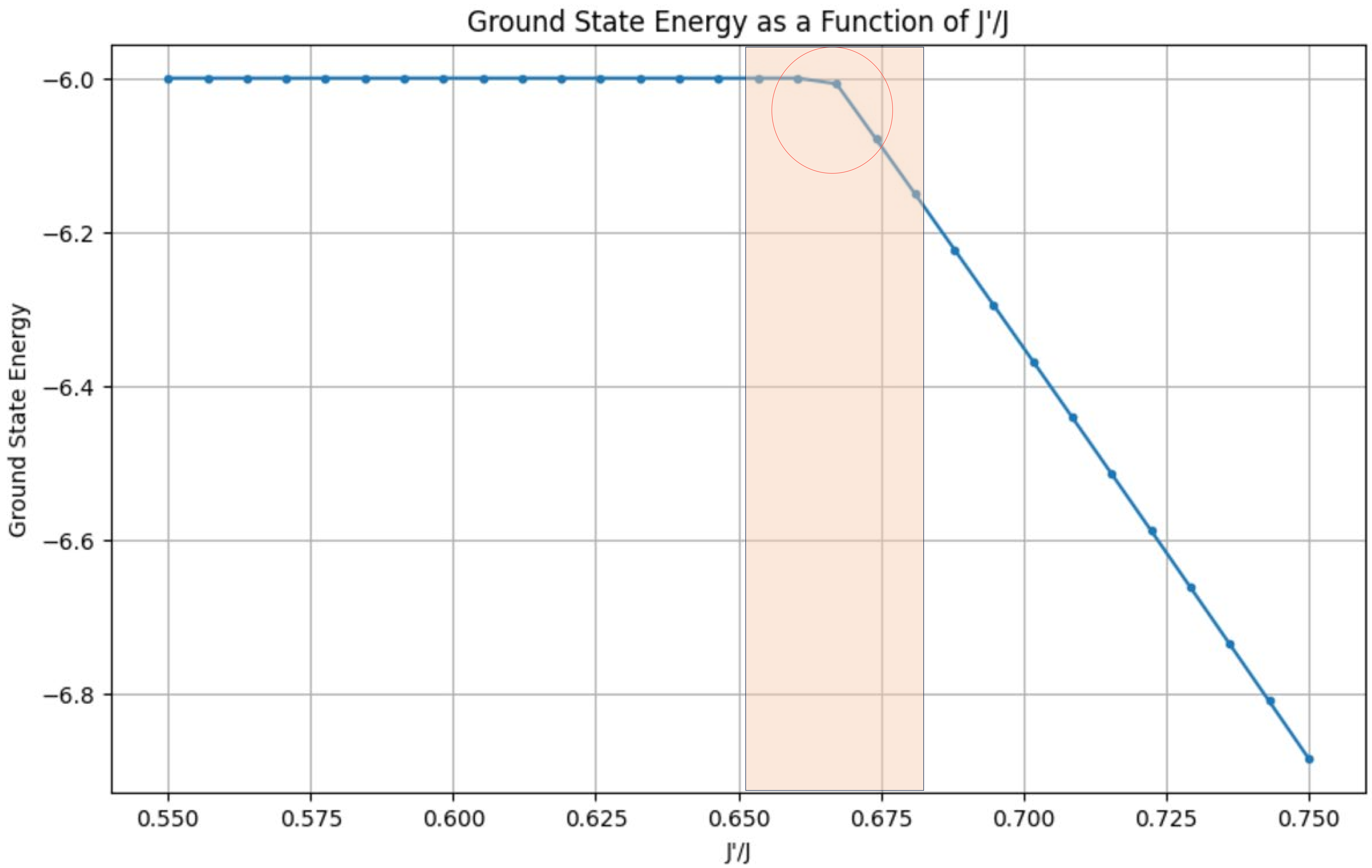
```
Possible Spins and Occurrences:
Spin: 0.0, Occurrence: 1430
Spin: 1.0, Occurrence: 3432
Spin: 2.0, Occurrence: 3640
Spin: 3.0, Occurrence: 2548
Spin: 4.0, Occurrence: 1260
Spin: 5.0, Occurrence: 440
Spin: 6.0, Occurrence: 104
Spin: 7.0, Occurrence: 15
Spin: 8.0, Occurrence: 1

Final Spins and Degeneracies:
Spin       Degeneracy  Degeneracy per occurrence
0.0        1430.0      1.0
1.0        10296.0     3.0
2.0        18200.0     5.0
3.0        17836.0     7.0
4.0        11340.0     9.0
5.0        4840.0      11.0
6.0        1352.0      13.0
7.0        225.0       15.0
8.0        17.0        17.0

Dimension of Hilbert space: 65536
Sum of degeneracies: 65536.0
Degeneracies sum up correctly.
Number of singlets (J=0): 1430
```

Directly counting

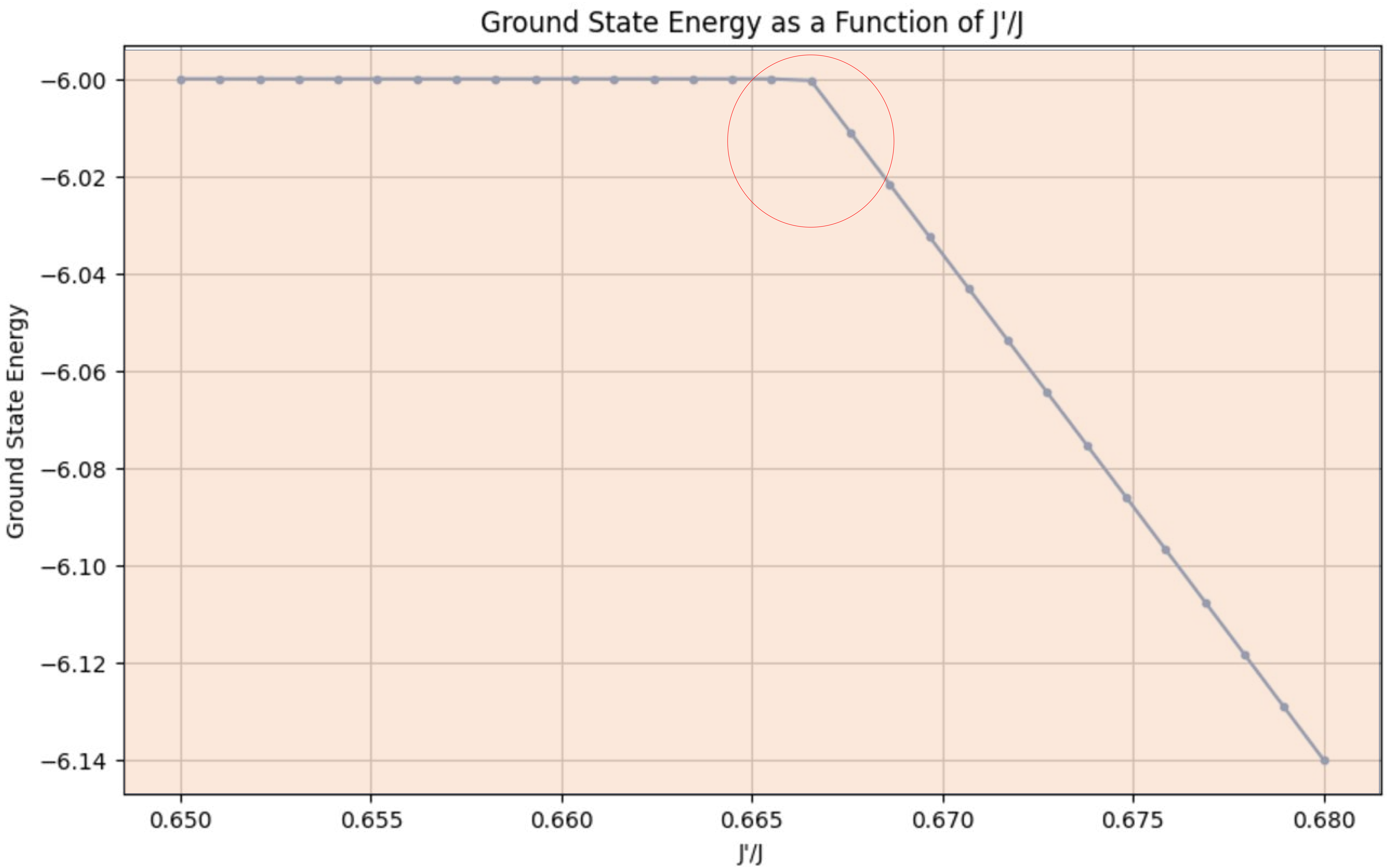
```
Unique spins and their counts:
Spin 0: 1430 occurrences
Spin 1: 3432 occurrences
Spin 2: 3640 occurrences
Spin 3: 2548 occurrences
Spin 4: 1260 occurrences
Spin 5: 440 occurrences
Spin 6: 104 occurrences
Spin 7: 15 occurrences
Spin 8: 1 occurrences
```

Obtained from the S^2_{ssm} matrices.

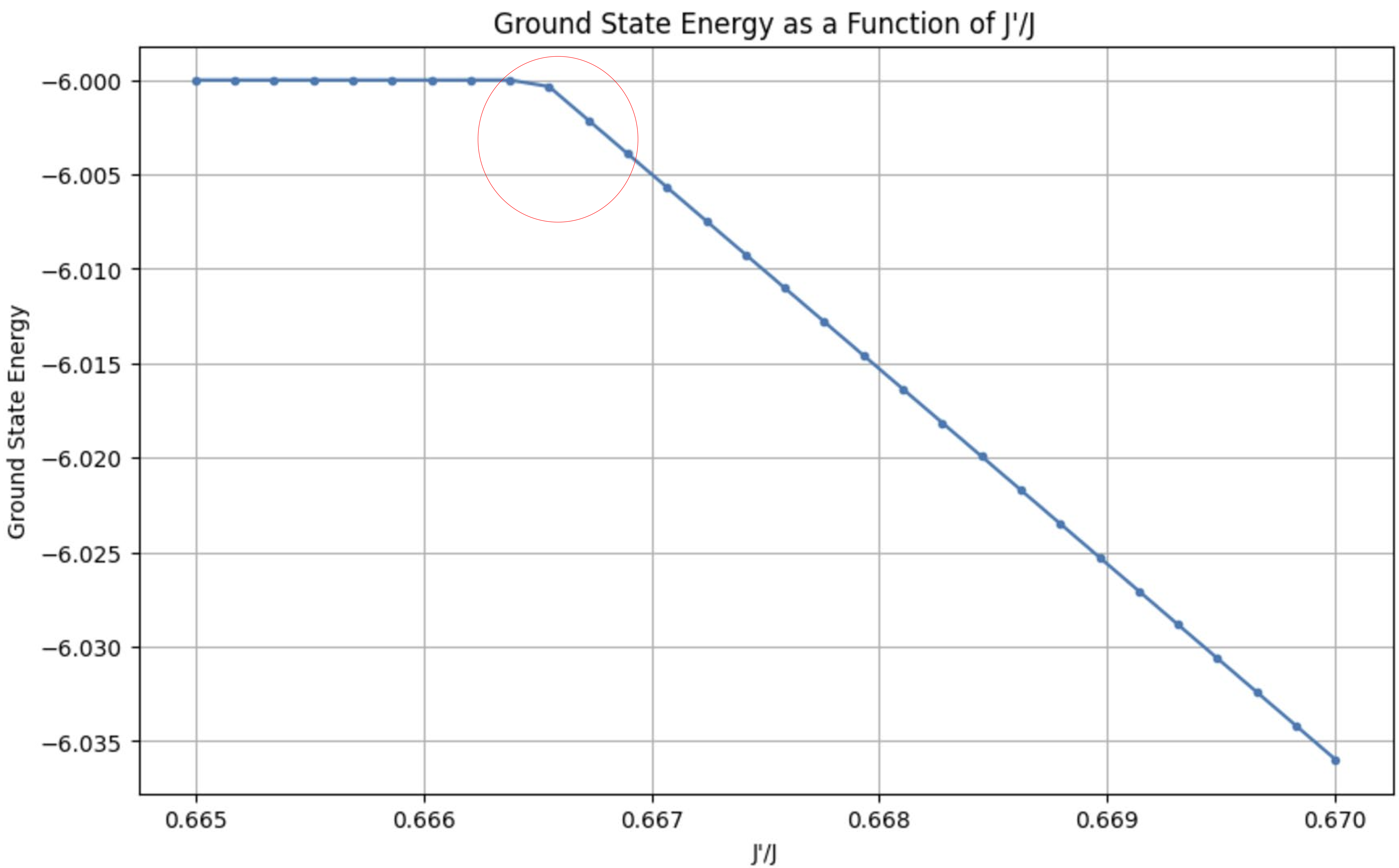
Duke Compute Cluster (DCC): Improved Result Granularity



Duke Compute Cluster (DCC): Improved Result Granularity

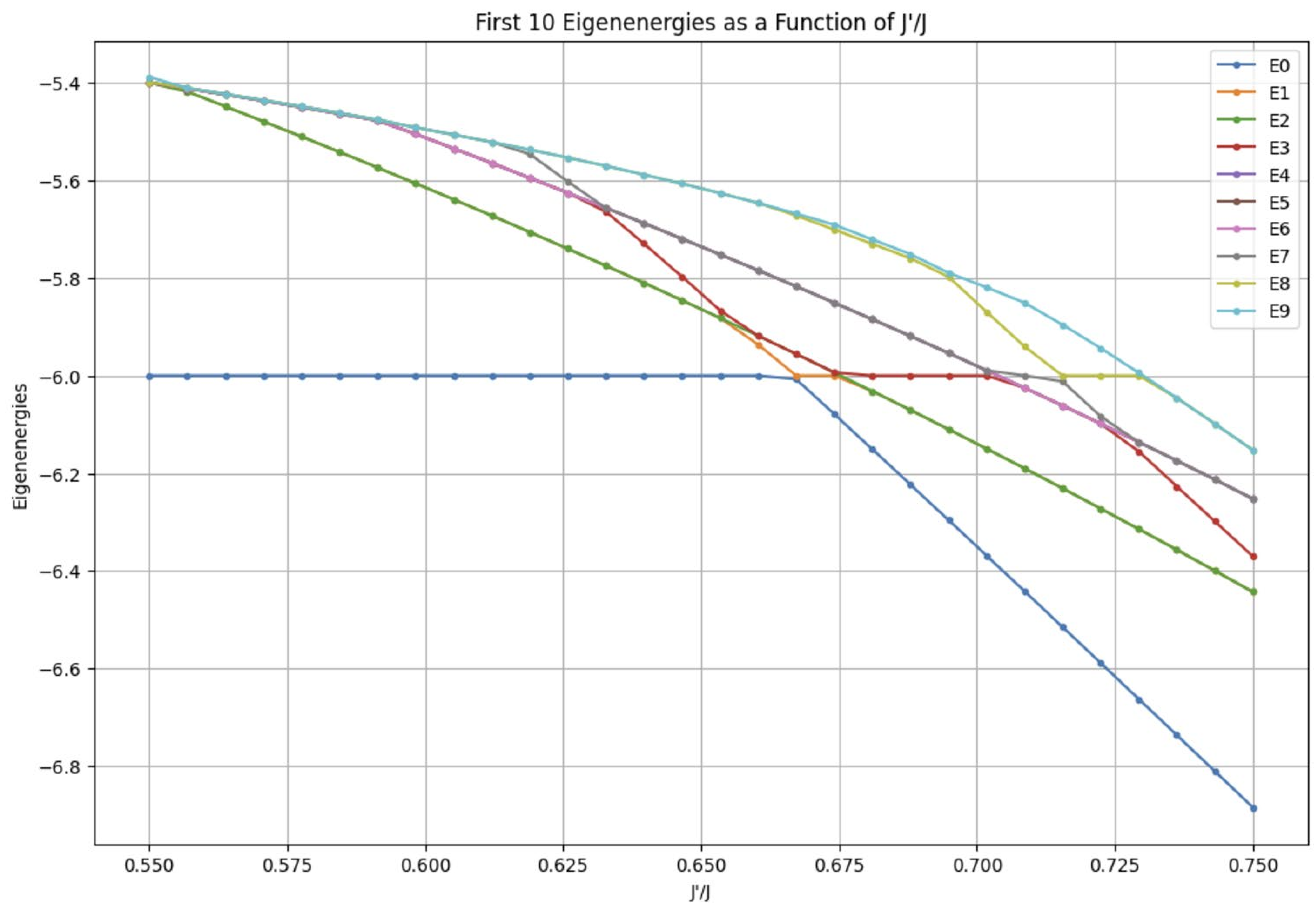


Duke Compute Cluster (DCC): Improved Result Granularity



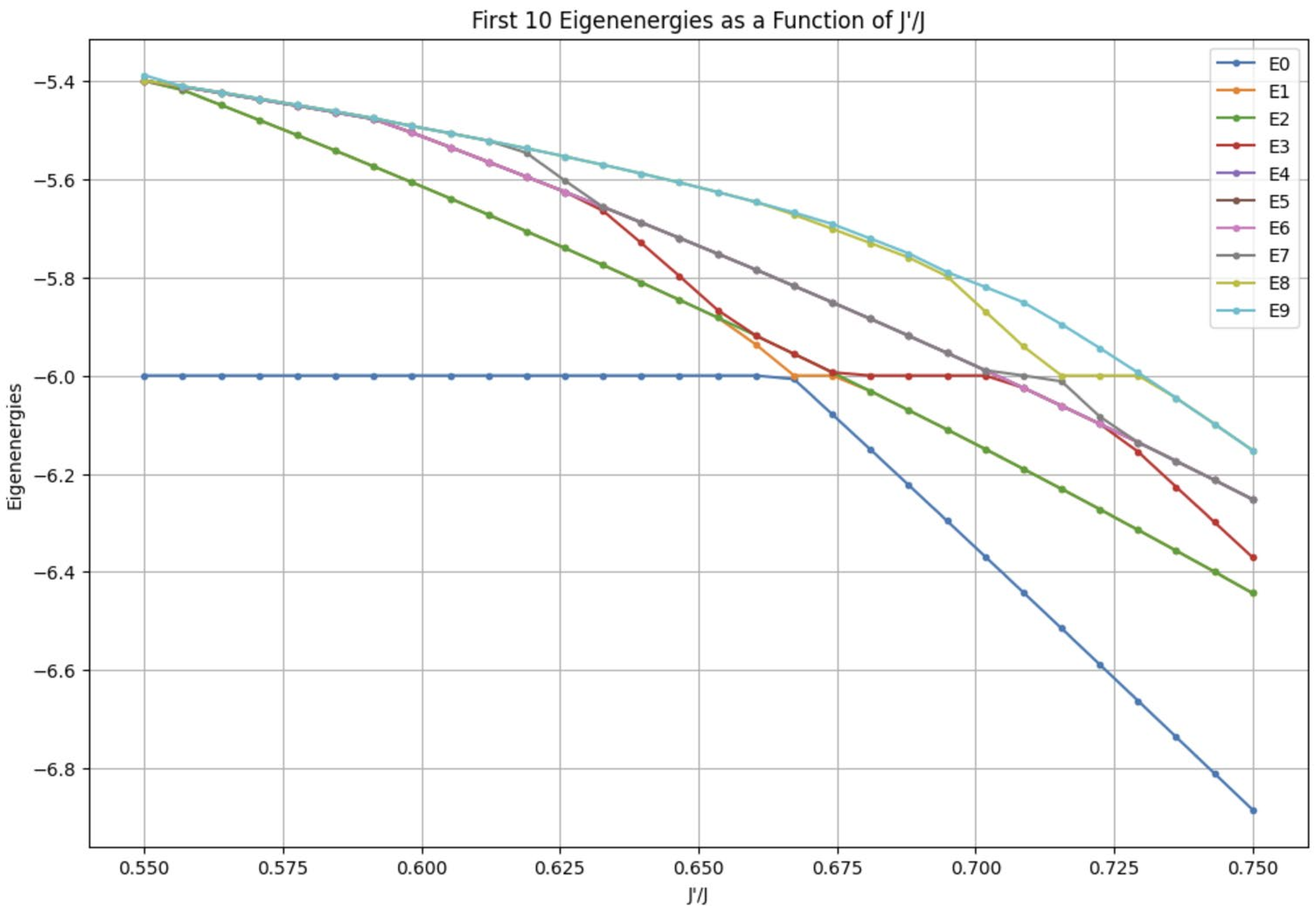
Energy Crossing Diagram: First Ten States

- Now, we have all the excited states too. So, we can plot the first ten eigenenergies and eyeball the crossing through (almost) adiabatic approximation.



- It looks like the first excited state becomes the ground state. However, we can obtain the total spin *s* value at 0.66 close to the first crossing.

What are the crossing states? Singlets or Triplets?



- We can obtain the total spin *s* value at 0.66 close to the first crossing.

- The first 8 states are all singlets! Hence, we know that out of the 1430 singlets, some come down to replace the dimer product as the ground state. Related to Quantum Spin Liquid (QSL).

```
Eigenvalue: -6.0
S^2 Matrix:
[[0]]

Eigenvalue: -5.3202867
S^2 Matrix:
[[2. 0 0 0]
 [0 2. 0 0]
 [0 0 2. 0]
 [0 0 0 2.]]

Eigenvalue: -5.3200417
S^2 Matrix:
[[2. 0 0 0]
 [0 2. 0 0]
 [0 0 2. 0]
 [0 0 0 2.]]

Eigenvalue: -5.1826
S^2 Matrix:
[[0 0]
 [0 0]]

Eigenvalue: -5.1172688
S^2 Matrix:
[[0 0 0 0]
 [0 0 0 0]
 [0 0 0 0]
 [0 0 0 0]]

Eigenvalue: -5.0296247
S^2 Matrix:
[[0]]

Eigenvalue: -5.0270248
S^2 Matrix:
[[0]]

Eigenvalue: -5.0261029
S^2 Matrix:
[[0 0]
 [0 0]]
```

J'/J = 0.5

```
Eigenvalue: -6.0
S^2 Matrix:
[[0.]]

Eigenvalue: -5.9330962
S^2 Matrix:
[[0.]]

Eigenvalue: -5.9167996
S^2 Matrix:
[[0. 0.]
 [0. 0.]]

Eigenvalue: -5.7828101
S^2 Matrix:
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]

Eigenvalue: -5.6456215
S^2 Matrix:
[[2. 0. 0. 0.]
 [0. 2. 0. 0.]
 [0. 0. 2. 0.]
 [0. 0. 0. 2.]]

Eigenvalue: -5.6422815
S^2 Matrix:
[[0.]]

Eigenvalue: -5.6393542
S^2 Matrix:
[[2. 0. 0. 0.]
 [0. 2. 0. 0.]
 [0. 0. 2. 0.]
 [0. 0. 0. 2.]]
```

J'/J = 0.66

Ideas for Future Works

- Explore the magnetization plateau theory of the SSL compounds.
- Implement a Quantum Monte Carlo on a bigger lattice and try to solve the sign problem.
- Understand the connection between SSM and Altermagnetism, a new phase of magnetism! Super new.
- Suggestions? Questions?